COE 158 Introduction to IT

Introduction to Information Technology

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Introduction…

“Man is still the most extraordinary computer of all”

John F. Kennedy

Lecture Outline

- What is Information Technology?
- Who Studies It?
- IT Infrastructure
  - Computers
  - Software
  - Users
  - Our View Today

IT Roles

Role	Job/Tasks
System Administrator	Administer the computers in an organization; install software; modify/update operating system; create accounts; train users; secure system; troubleshoot system; add hardware
Network Administrator	Purchase, configure and connect computer network; maintain computer network; troubleshoot network; secure network from intrusion
Database Administrator	Install, configure and maintain database and database management system; back up database; create accounts; train users
Web Administrator	Install, configure and maintain web site through web server; secure web site; work with developers
Web Developer	Design and create web pages and scripts for web pages; maintain web sites
Security Administrator	Install, configure and administer firewall; create security policies; troubleshoot computer system (including network); work proactively against intrusions

System Administration

- Account management
- Password management
- File protection management
- Installing and configuring new hardware
- Troubleshooting hardware
- Installing and configuring software
- Updating software
- Providing documentation and support
- System-level programming (scripting)
- System-level security
- IT Support
  - Documentation, manuals, website support (wikis, FAQs), A/V demos
  - Training
  - Help desk support

Skills for the Successful IT Person

Skill	Description	Example(s)
Troubleshooting, problem solving	• Detect a problem • Diagnose its cause • Find a solution (means of fixing it)	• Poor processor performance • Disk space full • Virus or Trojan horse infection
Knowledge of operating systems	• Operating system installation • Application software installation • User account creation • System monitoring	• Versions of Linux • Versions of Unix • Windows • Mac OS
System level programming	• Shell scripts to automate processes • Manipulating configuration files for system services	• Bash, Csh scripts • DOS scripts • Ruby scripts • C/C++ programs

Continued

Skill	Description	Example(s)
System security	• Ensuring proper system security is in place • Following or drafting policies for users • Monitoring for threats	• Configuring a system firewall • Installing antiviral/anti-malware software • Examining log files for evidence of intrusion and system security holes • Keeping up with the latest security patches
Hardware	• Installing and configuring new hardware • Troubleshooting, replacing or repairing defective hardware	• Replacing CPUs and disk drives • Connecting network cables to network hubs, switches, routers

Be Prepared

- Keep up with new trends in technology
  - Be a self-starter
  - Be curious
  - Be willing to learn on your own
- You might be on call 24/7
- Behave ethically at all times
  - Know the difference between ethical and unethical behavior

Continued

What is Information ?

Continued

- Information is any knowledge that can be communicated, including abstract ideas and concepts

- The objects we represent can be either physical objects from the real world, such as products in an inventory, abstract objects e.g. mathematical construct

Continued

- **What is Technology ?**

  - Technology is application of scientific knowledge for practical purpose

  - Using scientific knowledge to store, retrieve and manipulate data using computers

In your own words what is information technology?

Continued

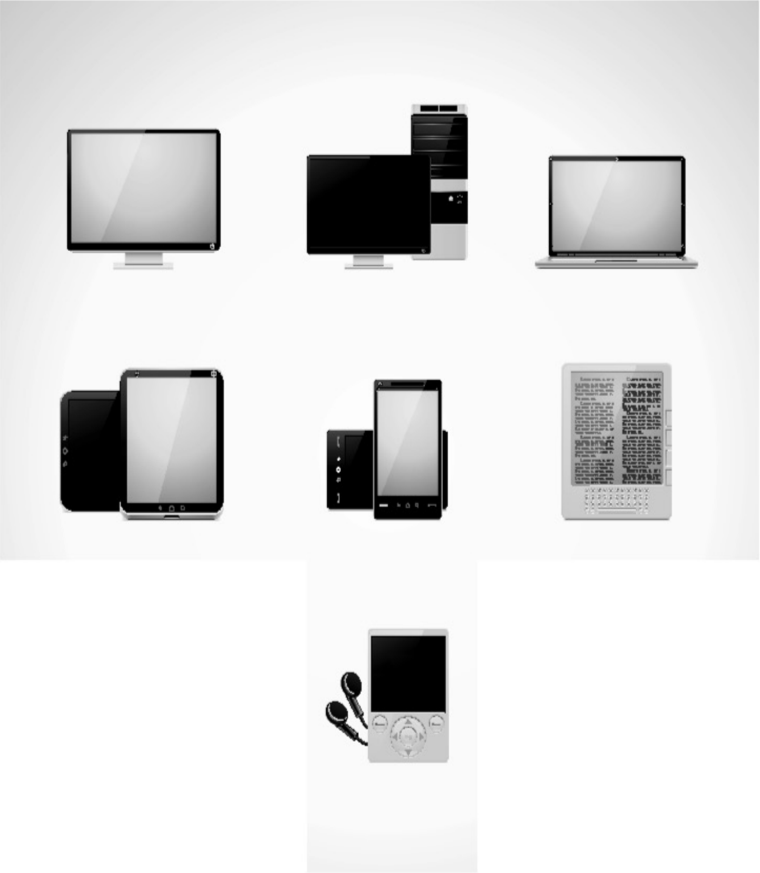
- **What is data ?**
  - Data is information in a form that the computer can use, for example, numbers, letters, and symbols.
  - Data come in many different forms: letters, words, integer numbers, real numbers, dates, times, coordinates on a map, and so on.
  - Data is unprocessed information

What is a Computer?

- Electronic device
  - General purpose (executes any type of program)
  - Stores data
  - Performs the Input-Processing-Output-Storage (IPOS) Cycle
  - A programmable device that can store, retrieve, and process data.

- Examples:
  - Supercomputers
  - Mainframes/minicomputers
  - Personal computers/laptops
  - Notebook computers
  - Handheld devices

- What about these
  - Smart phones
  - Mp3 players
  - Electronic book readers
  - GPS



Components of a Computer

- Processor
- Storage
  - Short term (memory)
    - SRAM
    - DRAM
    - ROM
  - Long term
    - Hard disk
    - Optical disk
    - Floppy disk
    - Flash drive
    - Magnetic tape
- I/O devices

- The system unit
- The motherboard, which contains
  - The CPU
  - CPU cooling unit
  - Extra processors (optional)
  - Memory chips for RAM, ROM
  - Connectors (ports) for peripherals
  - Expansion slots for peripheral device cards
  - ROM BIOS (booting, basic input & output instructions)
  - Power supply connector
- Disk drives
- Fan units
- Power supply

Storage Sizes

Size	Meaning	Example
1 bit	A single 0 or 1	Smallest unit of storage, might store 1 black & white pixel or 1 true/false value, usually we have to combine many bits to create anything meaningful
1 byte (1B)	8 bits	We might store a number from 0 to 255 or -128 to 127, or a single character (letter of the alphabet, digit, punctuation mark)
1 word	32 or 64 bits	One piece of data such as a number or a program instruction
1KB	1024 bytes	We might store a block of memory in this size
1MB	~1 million bytes	A small image or a large text file, an mp3 file of a song might take between 3 and 10 MB, a 50 minute tv show highly compressed might take 350MB
1GB	~1 billion bytes	A library of songs or images, dozens of books, DVD requires several GB of storage (4-8GB)
1TB	~1 trillion bytes	A library of movies

I/O Devices (Peripherals)

Software

- Programs
- Step-by-step instructions to the computer
- To run the program, the program must be written in the computer's machine language
- We write programs in high level languages
  - C/C++, Java, Python, Ruby, C#, Visual Basic, etc
- We translate programs from high level to machine language using language translator programs
  - compilers, interpreters, assemblers

Example Program

- Compare 2 numbers and output the larger
- In English:

  Input number1
  Input number2
  Compare the two numbers
  If the first is greater
      than the second
  Then output number1
  Otherwise output number2

- In C code:

```
scanf("%d", &number1);
scanf("%d", &number2);
if(number1 > number2)
    printf("%d is greater", number1);
else printf("%d is greater", number2);
```

Users

- Initiates the processes that computers run
- Uses interactivity
  - Makes computer usage more enjoyable
- Early users were engineers and programmers of the computers
- That changed in the 60s
- Today, nearly everyone is a user

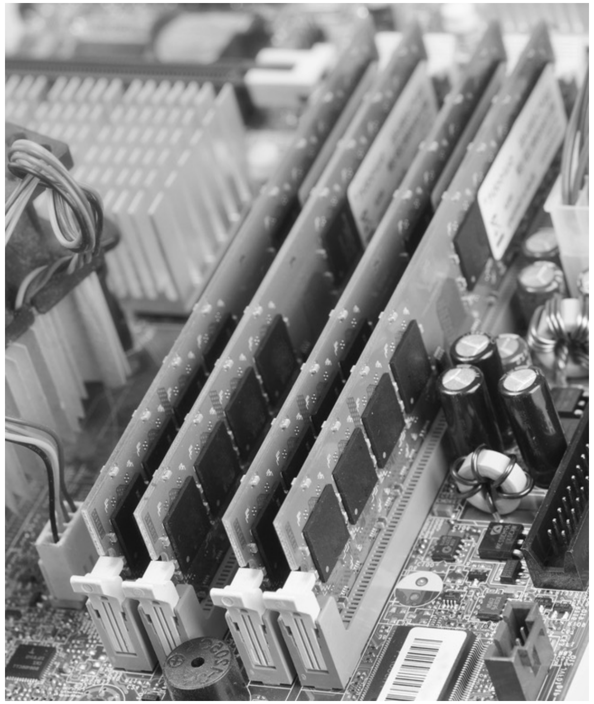
Computers Today

- Computers were easily identifiable
  - Expensive
  - Required clean room environments
  - People connected to them via network and terminals
- Not so today
  - Computers of any size
  - Wireless provides Internet access
  - Go anywhere and still have access
  - Billions of processors in our world

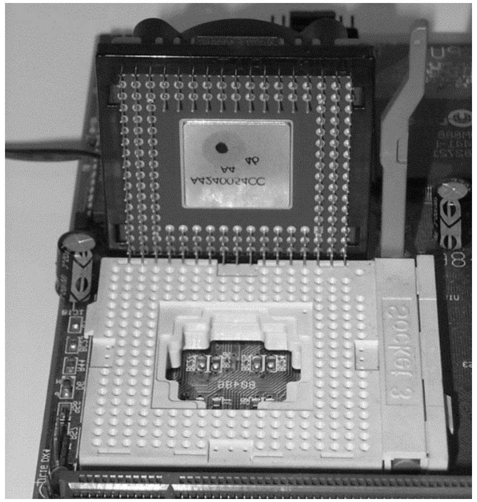
5MB Computer in 1956

500GB Computer

Computer Hardware



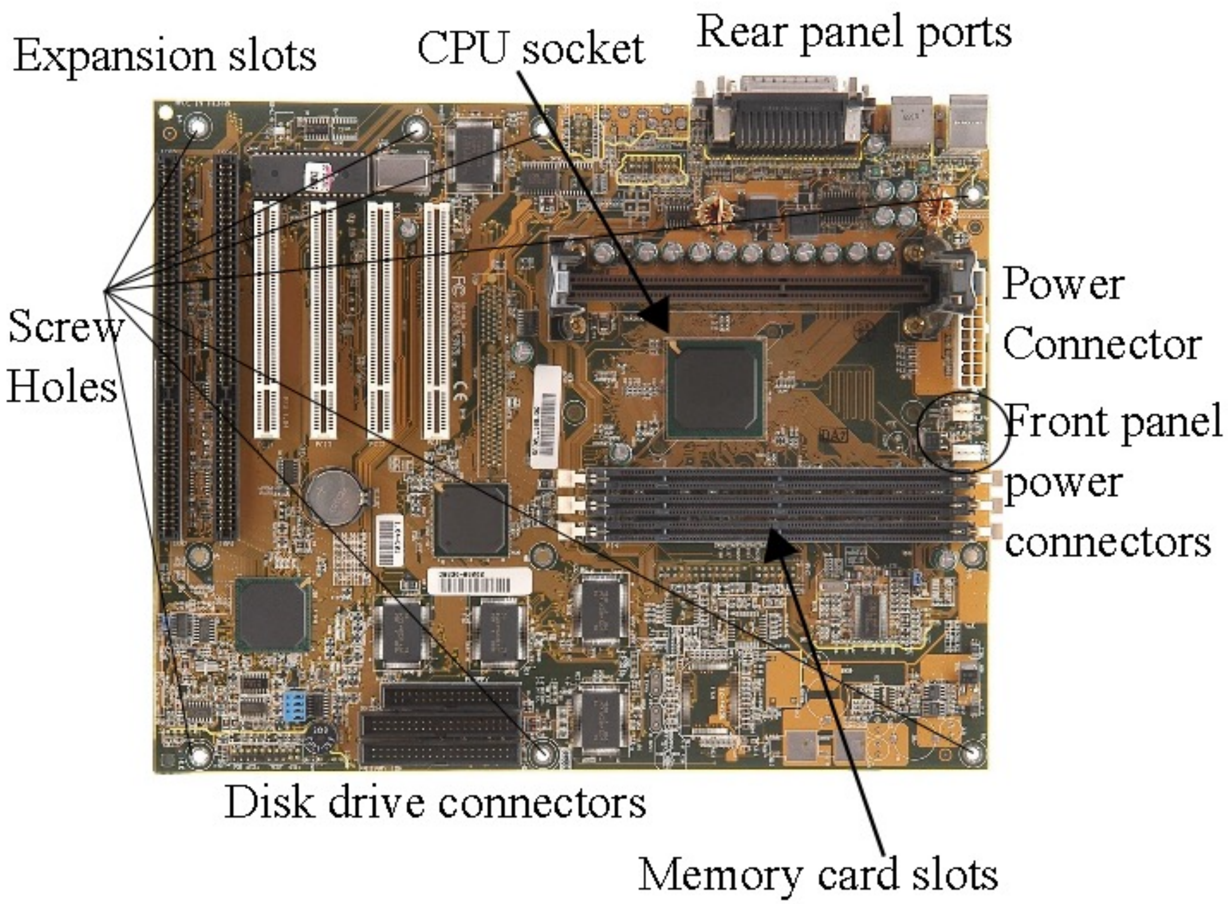
- Microprocessor – the CPU on 1 chip
  - Pins on one side to plug into motherboard CPU slot
  - Cooling unit sits on top
- Memory – individual chips placed on circuit board
  - Slip circuit board into memory slot(s)
- System bus – part of the motherboard



System Unit

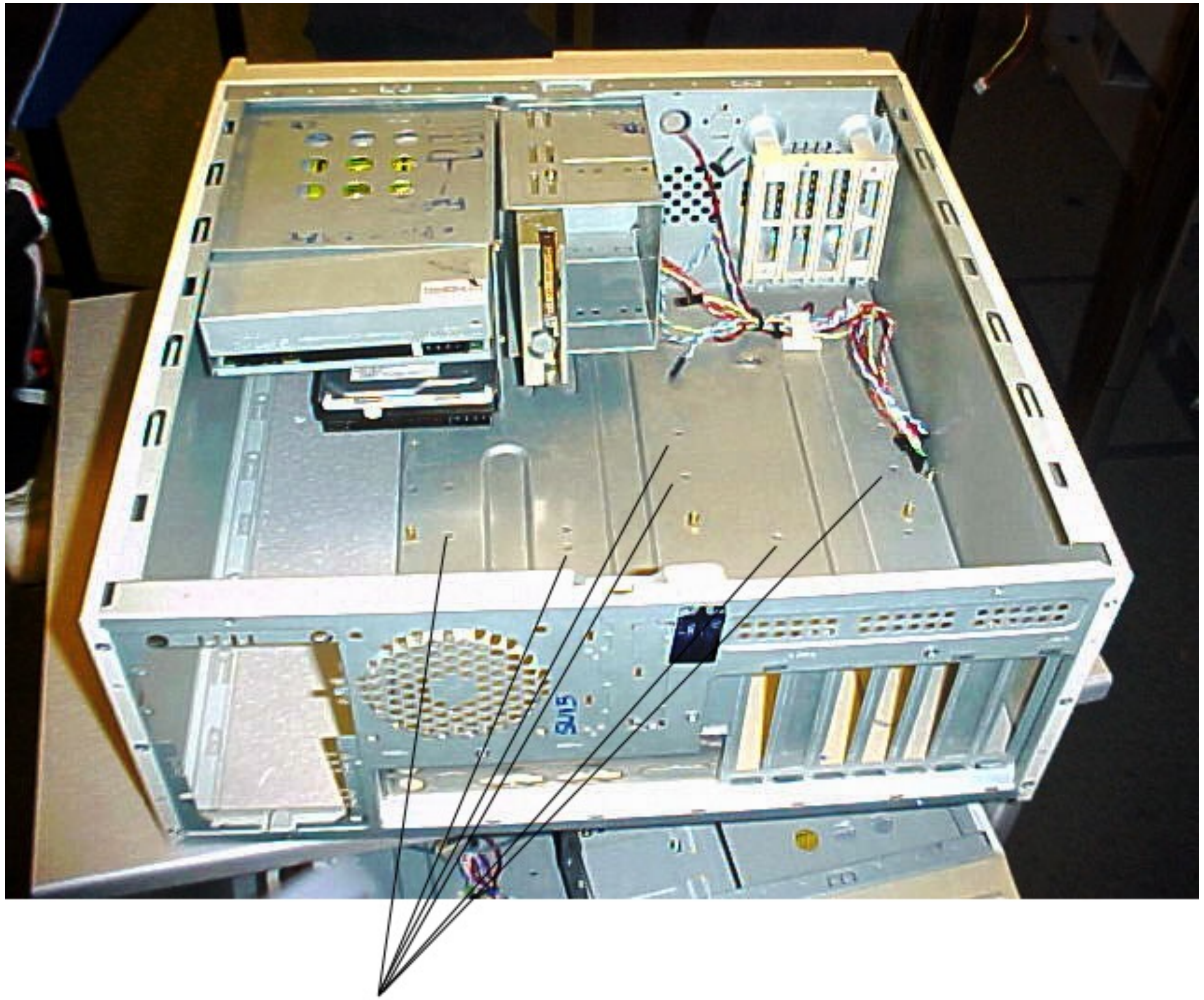
- Case into which
  - You place the motherboard
  - Fan(s)
  - Disk drive units
  - Power supply
  - Internal MODEM connected to motherboard directly
- Peripheral devices connect to motherboard through ports in system unit

Motherboard Explained



System Unit

Insert motherboard to rest on top of standoffs



Screw holes for standoffs

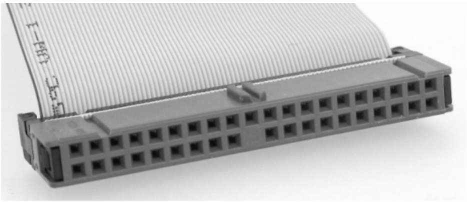
Hard Drive

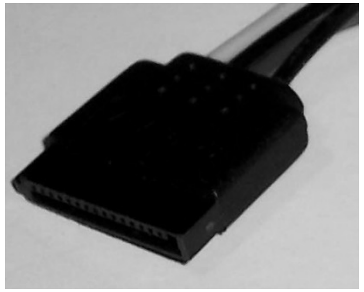
Multiple platters

1 read/write head
per surface

All disks spin in unison
by central spindle

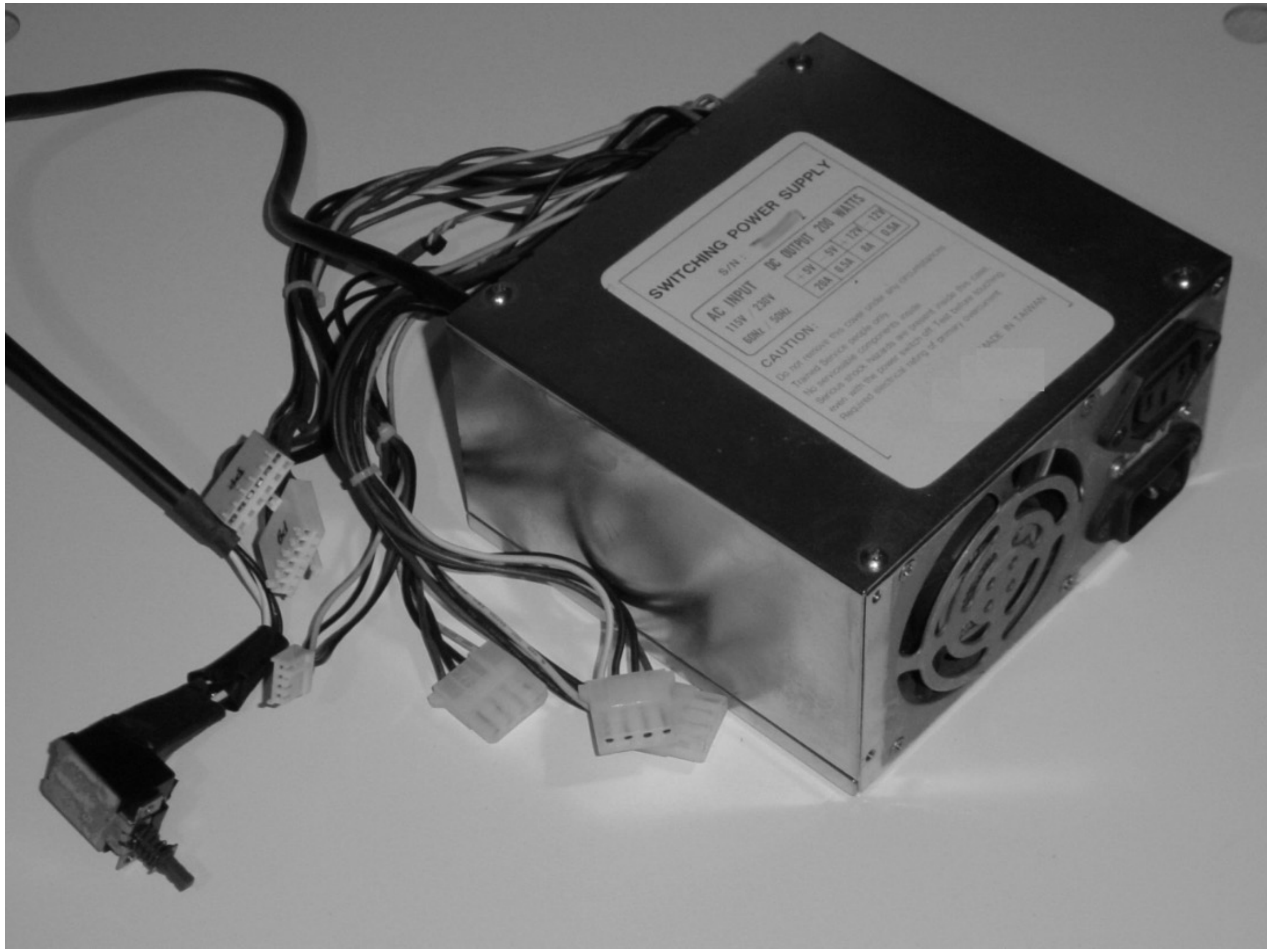
IDE (below) and SATA (below
to the right) use different
connectors



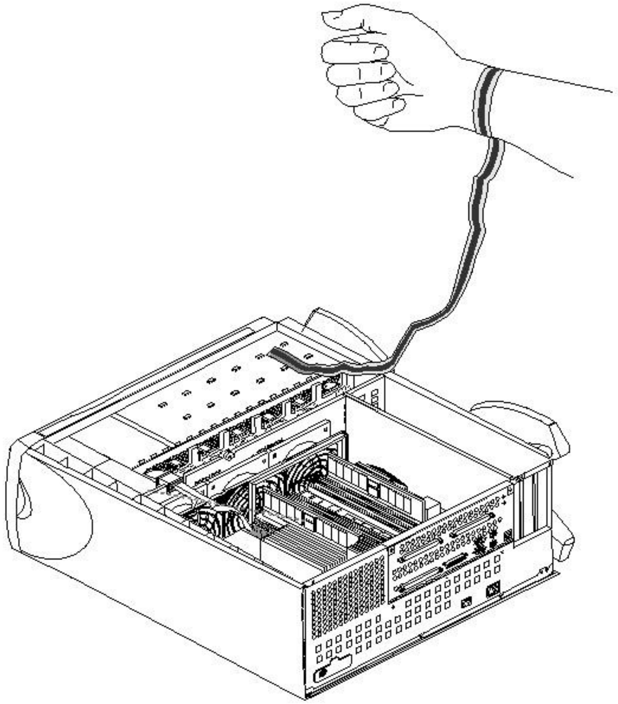


Power Supply

Cables connect to drives, fan and motherboard

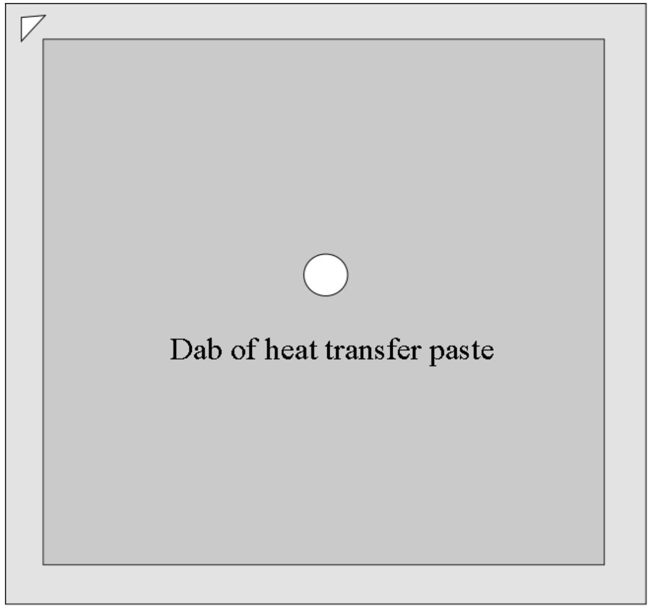


Computer Assembly



- Prepare system unit
  - Identify the components to assemble
  - Remove system unit side panel
  - Screw in standoffs
- Prepare motherboard
  - Wear grounding strap
  - Insert CPU
  - Use a dab of thermal energy paste
  - Attach cooling unit to CPU and plug in to motherboard
  - Attach memory circuit board(s)

CPU Cooling Unit



Place a "pea-sized" dab of heat paste on the surface of the CPU

Cooling unit "bolts" inserted into holes around CPU

Plug cooling unit's power connector into motherboard socket

Finish System Unit

- Insert disk drive unit(s)
  - Remove from bezel panel for optical drive but not for hard disk drive(s)
- Insert power supply unit into system unit
- Connect power supply cables to
  - Hard disk drive(s)
  - Optical disk drive
  - Motherboard LED panel
  - Fan
- Plug in external power cord to power supply unit, power on computer, does it boot?
  - When done, reinsert system unit panel

Insert Disk Drive(s) Into Slot(s)

- Disk drive bays may have rails otherwise slip rails to sides of drive(s)
- Slide disk drive into empty bay in system unit
- Best to place drives as far toward the top of the system unit as available